GATE
CS & DA

Artificial Intelligence

Proposition & Predictive Logic

Lecture No. 02

Recap of Previous Lecture

Topic • Argument (Pattern)

Topic • Rule of Inference.

Topic

Topic

Topic

Turn on Slide map

Topics to be Covered

Topic

Topic

Topic

Topic

Topic

- Rules of Inference
- Questions

Today is your opportunity to build the tomorrow you want.
—KEN POIROT
RD

Argument

- The statement that A set of premises (P1,P2,…...,Pn) yields another propostion Q is called an Argument.
- Q is called the conclusion of the Argument.
- So this is trying to say that when P1,P2,P3 ….. Pn are true then Q is also true….

Pattern

Argument

Example-1 :

- Every student of Information Technology studies Data Structures.
- Data structures necessarily contain the study of arguments.
- Therefore, every student of Information Technology studies arguments.

Argument

- Example-2 :

- Every parent is a mature person.
- Children should listen to mature people.
- Therefore, every child should listen to their parents.

Argument

- Example-3 :

- Every mother is a woman.
- All women are caring.
- Therefore, every mother is caring.

Argument

- Types of Arguments
- 1. Valid and Invalid Arguments
- Valid Argument: An argument is valid if the conclusion logically follows from the premises. In other words, if all the premises are true, the conclusion must be true as well.
- Validity is about the logical structure of the argument, not the actual truth of the premises or the conclusion.

Argument

- Premise 1: All humans are mortal.
  Premise 2: Socrates is a human.
  Conclusion: Socrates is mortal.
  In this example, if the premises are true, the conclusion must also be true, so the argument is valid.

Argument

- Invalid Argument: An argument is invalid if the conclusion does not logically follow from the premises, even if the premises are true. An invalid argument has a flaw in its logical structure, meaning the conclusion could be false even if the premises are true.

Argument

- Premise 1: If it rains, the ground is wet.
  Premise 2: The ground is wet.
  Conclusion: It is raining.
  This argument is invalid because the ground could be wet for reasons other than rain (e.g., someone watered the garden).

P	Q	P→Q
F	F	T
F	T	T
T	F	F
T	T	T

Critical Rows

Argument

- Methods of checking whether an argument is valid or not
- 1. Make truth table and make all columns of the different premises p1,p2,…….pn and also make a column of Q.
- 2. Mark the rows in which all of p1,p2,……..pn (i.e all premises) are true. such rows are called critical rows.
- 3. Then, in the critical rows, check the value of conclusion 'Q', if Q is true in all the critical rows, then the conclusion is valid
- Note: 1. There can be more than one critical row.
- 2. If there is anyone critical row in which Q is false, then the argument is invalid.
- 3. For the argument to be valid, 'Q' should be confirmed in all the critical rows in the truth table.

Argument

- Method 2: (Tautology Method)
- Just make a truth table in which make the columns of p1,p2,p3.....pn, Q , p1∧p2∧p3.....pn and finally $\boxed{p1 \wedge p2 \wedge p3 \wedge \ldots\ldots pn \rightarrow Q.}$ ← Tautology✓.
- if p1∧p2∧.....pn -> Q is a Tautology, then the argument is valid.

Rule of Inference

- A rule of inference is a logical principle that defines how new statements (conclusions) can be derived from existing statements (premises) in a valid way. These rules are the foundation of logical reasoning, ensuring that conclusions follow logically from premises.

Rule of Inference

- Modus Ponens (MP):
- Also known as affirming the antecedent, this rule allows you to infer the consequent if the antecedent is true.
- Form: $P \Longrightarrow Q$
- P (Premise P is true)
- Conclusion: Q (Therefore, Q is true)

P_1: $P \rightarrow Q$
P_2: P
Conclusion Q

Rule of Inference

- Example:
  Premise 1: If it rains, the streets will be wet.
- Premise 2: It is raining.
- Conclusion: The streets will be wet.

Rule of Inference

- Modus Tollens (MT):
- Also known as denying the consequent, this rule states that if the consequent is false, the antecedent must also be false.
- Form: $P \implies Q$ (If P, then Q)
- $\neg Q$ (Not Q)
- Conclusion: ¬P (Not P)

P_1: $P \rightarrow Q$

P_2: $\neg Q$

Conclusion = $\neg P$

Rule of Inference

- Example:
- Premise 1: If it rains, the streets will be wet.
- Premise 2: The streets are not wet.
- Conclusion: It is not raining.

Rule of Inference

- Hypothetical Syllogism (HS):
- This rule allows you to infer a direct relationship between two statements if one statement leads to another and that second statement leads to a third.
- Form: $P \Longrightarrow Q$
- $Q \Longrightarrow R$
- Conclusion: $P \Longrightarrow R$

P_1 $A \Rightarrow B$
P_2 $B \Rightarrow C$
then $A \Rightarrow C$

Rule of Inference

- Example:
- Premise 1: If I study, I will pass the exam.
- Premise 2: If I pass the exam, I will graduate.
- Conclusion: If I study, I will graduate.

Rule of Inference

4. Disjunctive Syllogism (DS):

- This rule is used when one of two possibilities is true, and one of them is false, allowing you to infer the other must be true.
- Form: $P \vee Q$ (Either P or Q)
- $\neg P$ (Not P)
- Conclusion: Q (Therefore, Q)

P_1 $A \vee B$
P_2 $\neg A$
Concl B

P_1 $A \vee B$
P_2 $\neg B$
Concl A

Rule of Inference

- Example: A: I will have Tea B: I will have coffee
- Premise 1: Either I will have coffee or tea. $(B \vee A)$
- Premise 2: I will not have coffee. $(\neg B)$
- Conclusion: I will have tea.

Cond.

Valid argument.

A	B	$\neg B$	$(B \vee A)$
F	F	T	F
F	T	F	T
T	F	T	T
T	T	F	T

Rule of Inference

- Addition:
- This rule allows you to add a disjunction (logical OR) with any statement.
- Form: P
- Conclusion: $P \vee Q$

(Premise A
Concl. $A \vee B$) Valid arg.

Rule of Inference

- Example:
- Premise: The sun is shining. (A)
- Conclusion: The sun is shining, or it is raining. (A ∨ B)

Valid

B	A	A∨B
F	F	F
F	T	T ✓
T	F	T
T	T	T ✓

Premise Concl.

Rule of Inference

- Simplification: This rule allows you to infer one part of a conjunction.
- Form: $P \wedge Q$ (Both P and Q are true)
- Conclusion: P

P_1: $A \wedge B$
Concl: A

Rule of Inference

- Example:
- Premise: I have a pen and a notebook.
- Conclusion: I have a pen.

A

B ⇒ I have a notebook

A

Concl. Premise

A	B	A ∧ B
F	F	F
F	T	F
T	F	F
T	T	T

Rule of Inference

- . Conjunction:
- This rule allows you to combine two true statements into a single conjunction.
- Form:P
- Q
- Conclusion: $P \wedge Q$

Rule of Inference

- Example:
- Premise 1: I have a pen. A
- Premise 2: I have a notebook. B
- Conclusion: I have a pen and a notebook. $A \wedge B$

A	B	$A \wedge B$
T	T	T

Rule of Inference

- Resolution:
- A rule used in disjunctive reasoning to infer new conclusions from disjunctions.
- Form: $P \vee Q$
- $\neg P \vee R$
- Conclusion: $Q \vee R$

P_1: $A \vee B$

P_2: $\neg A \vee C$

Conc: $B \vee C$

Rule of Inference

- Example:
- Premise 1: I will go for a walk (A) or stay home (B). (C)
- Premise 2: I will not go for a walk (¬A) or I will read a book (C).
- Conclusion: I will stay home (B) or read a book (C).

A	B	C	$A \vee B$	$(\neg A \vee C)$	$B \vee C$
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			

Valid.

A	B	C	P_1 $A \vee B$	P_2 $(\neg A \vee C)$	$B \vee C$
0	0	0	0	1	0
0	0	1	0	1	1
0	1	0	1	1	1
0	1	1	1	1	1
1	0	0	1	0	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	1	1

Rule of Inference

- Constructive Dilemma:
- This rule applies when two conditionals are true, and their disjunction implies a disjunction of their conclusions.

P_1 $(A \rightarrow B)$

P_2 $(C \rightarrow D)$

P_3 $A \vee C$

Conc $B \vee D$

Rule of Inference

- Example:
- Premise 1: If I study, I will pass. (P, Q)
- Premise 2: If I sleep well, I will feel rested. (R, S)
- Premise 3: Either I will study or sleep well. (P∨R)
- Conclusion: I will either pass or feel rested. (Q, S)

$P \to Q$ (F)
(1 0)

$R \to S$
1 0

$P \vee R$ (F)
0 0

$Q \vee S$ F
0 0

P	Q	R	S	$P \to Q$	$R \to S$	$P \vee R$	$Q \vee S$
1	0	0	0	8 F	2 F	0 F	0 F
1	0	0	1	9	6	1	2
1	0	1	0	10	10	4	8
1	0	1	1	11	14	5	10

0	0	1	0
0	1	1	0
1	0	1	0
1	1	1	0
0	0	0	0
0	0	0	1
0	1	0	0
0	1	0	1

0	0	0	0
0	0	1	0
1	0	0	0
1	0	1	0

~~0~~ ~~1~~ ~~2~~ 3 ~~4~~ ~~5~~ ~~6~~ 7 ~~8~~ ~~9~~ ~~10~~ ~~11~~ 12 13 ~~14~~ 15

- So all Premise are T @3,7,12,13,15
- and @ all these points Conclusion is also T So Valid

P	Q	R	S	P→Q	R→S	PvR	QvS
1	0	0	0	8 F	2 F	0 F	0 F
1	0	0	1	9	6	1	2
1	0	1	0	10	10	4	8
1	0	1	1	11	14	5	10

0 0	1 0
0 1	1 0
1 0	1 0
1 1	1 0
0 0	0 0
0 0	0 1
0 1	0 0
0 1	0 1

0	0 0 0
0	0 1 0
1	0 0 0
1	0 1 0

Q1 P: $A \wedge B$

Concl: A

$A \wedge B$ = True then $A = T$, $B = T$

Valid

Q2: P_1: $A \vee B$ P_2: $\neg B$

Concl: A

$A \vee B$ = True, $\neg B$ = True

$\Rightarrow B$ = false

Now Since $A \vee B$ = True, B = false

then A has to be True

Q3. P_1: $\neg(A \wedge B)$

P_2: A

Concl: $\neg B$

Valid

Conc

A	B	$\neg B$	$\neg(A \wedge B)$
0	0	1	1
0	1	0	1
1	0	1	1
1	1	0	0

Q. $P_1 : P \to R$

$P_2 : Q \to S$

$P_3 : \neg R \vee \neg S$

Concl. $\neg P \vee \neg R$

invalid ✓

P	Q	R	S	$P \to R$	$Q \to S$	$\neg R \vee \neg S$	$\neg P \vee \neg R$
1	0	0	0	F	F	F	F
1	0	0	1				
1	1	0	0				
1	1	0	1				
0	1	0	0				
0	1	1	0				
1	1	0	0				
1	1	1	0				
0	0	1	1				
0	1	1	1				
1	0	1	1				
1	1	1	1				

				8	4	3	10
				9	6	7	11
				12	12	11	14
				13	14	15	15

P	Q	R	S
1	0	1	0
1	0	1	1
1	1	1	0
1	1	1	1

0 1 2 ~~3~~ ~~4~~ 5 ~~6~~ ~~7~~

~~8~~ ~~9~~ 10 ~~11~~ ~~12~~ ~~13~~ ~~14~~ ~~15~~

0, 1, 2, 5, 10

[NAT]

Simplified from of the statement

$\{\sim p \wedge (\sim Q \wedge R)\} \vee \{(Q \wedge R) \vee (P \wedge R)\}$

$(\sim P \wedge (\sim Q \wedge R)) \vee ((Q \wedge R) \vee (P \wedge R))$

$\rightarrow (\sim P \wedge \sim Q \wedge R) \vee (Q \wedge R) \vee (P \wedge R)$

a) T

b) f

c) Q

d) R

P Q R	Result
0 0 0	0
0 0 1	1
0 1 0	0
0 1 1	1
1 0 0	0
1 0 1	1
1 1 0	1
1 1 1	1

P Q R
0 0 1
0 1 1
1 1 1
1 0 1
1 1 1

$$(\sim P \wedge (\sim Q \wedge R)) \vee ((Q \wedge R) \vee (P \wedge R))$$

a) T

b) F

c) Q

d) R

$$\underbrace{(\sim P \wedge \sim Q \wedge R)}_{1} \vee \underbrace{(Q \wedge R)}_{1} \vee \underbrace{(P \wedge R)}$$

PQR	
000	0
001	1
010	0
011	1
100	0
101	1
110	0
111	1

PQR
001
011
111
101
111

$$\left(\sim P \wedge (\sim Q \wedge R)\right) \vee \left((Q \wedge R) \vee (P \wedge R)\right)$$

$$\Downarrow \qquad\qquad \downarrow$$

$$(\sim P \wedge \sim Q \wedge R) \vee ((Q \vee P) \wedge R)$$

$$(\sim (P \vee Q) \wedge R) \vee ((P \vee Q) \wedge R)$$

$$\left(\sim (P \wedge Q) \vee (P \vee Q)\right) \wedge R$$

$$\Rightarrow T \wedge R \Rightarrow R$$

[NAT]

Simplified form of the statement

$[(P \vee Q) \wedge \sim\{\sim p \wedge (\sim Q \vee \sim R)\}] \vee [(\sim P \wedge \sim Q) \vee (\sim P \wedge \sim R)]$

P	Q	R
0	0	0
0	0	1
0	1	0 →
0	1	1 →
1	0	0 →
1	0	1 →
1	1	0 →
1	1	1 →

$(P \vee Q) \wedge \sim(\sim P \wedge \sim(Q \wedge R)) \vee [\sim(P \vee Q) \vee \sim(P \vee R)]$

$[(P \vee Q) \wedge (P \vee (Q \wedge R))] \vee [\sim(P \vee Q) \vee \sim(P \vee R)]$

(2,3,4,5,6,7) (3,4,5,6,7) (0,1) (0,2)

P	Q	R	
0	0	0	
0	0	1	
0	1	0	
0	1	1	→ 1
1	0	0	→ 1
1	0	1	→ 1
1	1	0	→ 1
1	1	1	→

(3,4,5,6,7)

(0,1,2)

P	Q	R	P∨Q	P∨R
0	0	0	0	0 ✓
0	0	1	0	1
0	1	0	1	0 ✓
0	1	1	1	1
1	0	0	1	1
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

So we get $1 @ (0,1,2,3,4,5,6,7)$

[MCQ]

Which of the following implication is/are valid

or

which of the following is/are a tautology

A $(P \wedge Q) \rightarrow (P \leftrightarrow Q)$

$A \rightarrow B$

$(P \wedge Q) \rightarrow (P \leftrightarrow Q)$

B $(P \leftrightarrow Q) \rightarrow (P \rightarrow Q)$

C $(P \leftrightarrow Q) \rightarrow (P \rightarrow \sim Q)$

D $Q \rightarrow (P \rightarrow Q)$

P	Q	A: $P \wedge Q$	B: $P \leftrightarrow Q$	$A \rightarrow B$
F	F	F	T	T
F	T	F	F	T
T	F	F	F	T
T	T	T	T	T

P	Q	A: $P \leftrightarrow Q$	B: $P \rightarrow Q$	$A \rightarrow B$
F	F	T	T	T
F	T	F	T	T
T	F	F	F	T
T	T	T	T	T

[MCQ]

Which of the following implication is/are valid

or

which of the following is/are a tautology

A. $(P \wedge Q) \rightarrow (P \leftrightarrow Q)$ ($A \rightarrow B$)

$(P \wedge Q) \rightarrow (P \leftrightarrow Q)$

B. $(P \leftrightarrow Q) \rightarrow (P \rightarrow Q)$

C. $(P \leftrightarrow Q) \rightarrow (P \rightarrow \sim Q)$

D. $Q \rightarrow (P \rightarrow Q)$

P	Q	A: $P \leftrightarrow Q$	B: $P \rightarrow \neg Q$	$A \rightarrow B$
F	F	T	T	T
F	T	F	T	T
T	F	F	T	T
T	T	T	F	F

P	Q	A: $P \rightarrow Q$	B: $A \rightarrow B$
F	F	T	T
F	T	T	T
T	F	F	T
T	T	T	T

[MCQ]

Which of the following is/are false

A. $a \rightarrow (b \vee c) \equiv (a \wedge \sim b) \rightarrow c$

B. $(P \rightarrow Q) \wedge (R \rightarrow Q) \equiv (P \vee R) \rightarrow Q$

C. $(P \rightarrow Q) \vee (R \rightarrow Q) \equiv (p \wedge R) \rightarrow Q$

[MCQ]

Which of the following arguments is/are valid

A. Conclusion R follows prom the premises $\{P \rightarrow (Q \rightarrow R), P \wedge Q\}$

B. $\{P \rightarrow (R \rightarrow S), \sim R \rightarrow \sim P, P\}$ logically implies 'S'

C. $(P \wedge Q)$ follows from the premises $\{(P \vee Q), \sim P\}$

D. $\{\sim T \rightarrow \sim R, \sim S, T \rightarrow W, R \vee S\} \mapsto W$

[NAT]

Check whether the following argument is valid or invalid

$\{P \rightarrow Q, \sim(P \wedge Q)\}$ yields $\sim P$

Proof by Contradiction

Proof by Contradiction

[NAT]

Check whether the argument is valid or invalid

$\{P \rightarrow (Q \rightarrow S), \sim R \vee P, Q\}$ logically implies $(R \rightarrow S)$

THANK - YOU